



Skills Assessment Test

IT Questions

1. Software is defined as _____

- a) set of programs, documentation & configuration of data
- b) set of programs
- c) documentation and configuration of data
- d) None of the mentioned

Answer: a

Explanation: Software is a collection of programmes; it also includes documentation and data setup to enable the programmes to function. Microsoft windows, excel, word, powerpoint, etc. are a few examples of software.

2. What is Software Engineering?

- a) Designing a software
- b) Testing a software
- c) Application of engineering principles to the design a software
- d) None of the above

Answer: c

Explanation: Software engineering is the application of engineering principles to the design, development, and support of software and it helps to solve the challenges of low-quality software projects.



3. Who is the father of Software Engineering?

- a) Margaret Hamilton
- b) Watts S. Humphrey
- c) Alan Turing
- d) Boris Beizer

Answer: b

Explanation: Watts S. Humphrey created the Software Process Program at Carnegie Mellon University's Institute (SEI) in the 1980s, and served as its director from 1986 through the early 1990s. This program was designed to help participants understand and manage the software development process.

4. What are the features of Software Code?

- a) Simplicity
- b) Accessibility
- c) Modularity
- d) All of the above

Answer: d

Explanation: Software code should be written in a clear, succinct, and easy-to-understand way. Simplicity should be preserved in the program code's organization, implementation, and design. These codes should be constructed in such a way that software components (such as files and functions) are readily available. The software may be broken down into numerous parts to make it easier to comprehend and troubleshoot.

5. _____ is a software development activity that is not a part of software processes.

- a) Validation
- b) Specification
- c) Development
- d) Dependence

Answer: d

Explanation: A software dependency is an external independent library that can range in size from a single file to numerous files and directories arranged into packages to accomplish a specified purpose and is an attribute and not an engineering activity for process.

6. Define Agile scrum methodology.

- a) project management that emphasizes incremental progress
- b) project management that emphasizes decremental progress
- c) project management that emphasizes neutral progress
- d) project management that emphasizes no progress

Answer: a

Explanation: Agile scrum methodology is a style of project management that emphasizes incremental progress. Each iteration is divided into two to four-week sprints, with the goal of completing the most important features first and delivering a possibly deliverable product at the end of each sprint.

7. CASE stands for

- a) Computer-Aided Software Engineering
- b) Control Aided Science and Engineering
- c) Cost Aided System Experiments
- d) None of the mentioned

Answer: a

Explanation: The CASE tool's purpose is to make the work of software development and maintenance easier and more reliable.

8. _____ is defined as the process of generating analysis and designing documents?

- a) Re-engineering
- b) Reverse engineering
- c) Software re-engineering
- d) Science and engineering

Answer: b

Explanation: The method of reverse engineering is used to uncover difficult, unknown, and hidden information about a software system.

9. The activity that distributes estimated effort across the planned project duration by allocating the effort to specific software developing tasks is

- a) Project scheduling
- b) Detailed schedule
- c) Macroscopic schedule
- d) None of the mentioned

Answer: a

Explanation: Software project scheduling is the process of allocating anticipated effort to specific software developing activities and distributing it across the project's intended length. A macroscopic schedule is created in the early phases of project planning.

10. What is a Functional Requirement?

- a) specifies the tasks the program must complete
- b) specifies the tasks the program should not complete
- c) specifies the tasks the program must not work
- d) All of the mentioned

Answer: a

Explanation: A functional requirement is a specification that describes a system or one of its components. It specifies the tasks the program must complete. A function is made up of three parts: inputs, behavior, and outputs.

11. Why do bugs and failures occur in software?

- a) Because of Developers
- b) Because of companies
- c) Because of both companies and Developers
- d) None of the mentioned

Answer: c

Explanation: Software businesses are in charge of establishing policies and creating a



working environment for software development, therefore they are a component of the software development process. Bugs from developers are nothing new.

12. Attributes of good software is _____

- a) Development
- b) Maintainability & functionality
- c) Functionality
- d) Maintainability

Answer: b

Explanation: Good software should provide the functionality and maintainability that are necessary. Software development is a must, not an option. Along with this software also additional attributes like usability, efficiency, reliability, accuracy, robustness, integrity, etc.

13. The Cleanroom philosophy was proposed by _____

- a) Linger
- b) Mills
- c) Dyer
- d) All of the Mentioned

Answer: d

Explanation: Mills, Dyer, and Linger initially suggested the Cleanroom concept in the 1980s. The main goal of the philosophy of cleanroom philosophy is to develop software with zero defects.

14. What does SDLC stands for?

- a) System Design Life Cycle
- b) Software Design Life Cycle
- c) Software Development Life Cycle
- d) System Development Life cycle

Answer: c

Explanation: The Software Development Life Cycle (SDLC) is a method for designing,



developing, and testing high-quality software. The software developed to meet or exceed customer expectations must have an SDLC designed to complete the software on time and on budget.

15. Who proposed the spiral model?

- a) Barry Boehm
- b) Pressman
- c) Royce
- d) IBM

Answer: a

Explanation: The spiral model by Boehm, is a software process model that combines prototyping's iterative characteristic with the linear sequential model's regulated and systematic elements. It implements the capability of quick production of new software versions.

16. _____ is not among the eight principles followed by the Software Code of Ethics and Professional Practice.

- a) PRODUCT
- b) ENVIRONMENT
- c) PUBLIC
- d) PROFESSION

Answer: b

Explanation: The rest are software ethical provisions; the environment does not focus on individual clauses or their importance in relation to the topic. The eight principles that should be followed by the Software code is:

1. PUBLIC
2. CLIENT AND EMPLOYER
3. PRODUCT
4. JUDGMENT
5. MANAGEMENT



- 6. PROFESSION
- 7. COLLEAGUES
- 8. SELF

17. Which of the following are CASE tools?

- a) Central Repository
- b) Integrated Case Tools
- c) Upper Case Tools
- d) All of the mentioned

Answer: d

Explanation: CASE tools can be divided into Central Repository, Integrated Case Tools, Upper Case Tools, and Lower Case Tools based on their use at a particular SDLC stage.

18. _____ suits the Manifesto for Agile Software Development.

- a) Customer collaboration
- b) Individuals and interactions
- c) Working software
- d) All of the mentioned

Answer: d

Explanation: None.

19. Software patch is defined as _____

- a) Daily or routine Fix
- b) Required or Critical Fix
- c) Emergency Fix
- d) None of the mentioned



Answer: c

Explanation: When a vulnerability is discovered, a software patch is applied to the outdated version as an emergency repair. To repair a vulnerability or flaw discovered after an application or software has been released is referred to as software patching.

20. _____ software development team has no permanent leader.

- a) Controlled Centralized (CC)
- b) Controlled decentralized (CD)
- c) Democratic decentralized (DD)
- d) None of the mentioned

Answer: c

Explanation: Team members Communicate among themselves horizontally. There is no permanent leader in the team and a team of software engineers coordinates among themselves temporarily to perform the task.



Math Aptitude Test — 30 minutes total

Instructions

- Time: 30 minutes. Write answers clearly and show work.
- Calculators allowed unless stated otherwise.
- Answer all questions. Partial credit for clear steps.

Questions

- 1.Work-rate (4 min) A team of 6 people completes a job in 8 days. How many days will 4 people take at the same rate? Show steps.
- 2.Simple interest (3 min) You place \$5,000 at 4% simple interest per year. How much interest is earned in 3 years?
- 3.Percentage / increase (3 min) An item's price increases from \$120 to \$150. What is the percent increase? Show calculation.
- 4.Ratio and proportion (4 min) Mix A:B in ratio 3:2 to make 25 liters of solution. How many liters of A and B are needed?
- 5.Algebra — linear equation (4 min) Solve for x: $3(x - 2) + 4 = 2(x + 5) - x$.
- 6.Sequences (3 min) Find the next number in the sequence: 2, 6, 12, 20, 30, __. Explain rule.
- 7.Probability (3 min) A bag contains 5 red, 3 blue, and 2 green marbles. One marble is drawn at random. What is the probability it is red or green?
- 8.Two-step percent problem (3 min) After a 20% discount on an item, a sales tax of 8% is applied to the discounted price. List the final price for an item with original price \$250. (Show steps.)
- 9.Geometry — area (4 min) A rectangle has a perimeter of 60 cm and length is twice the width. Find the area.
- 10.Data interpretation (5 min) Table: Sales by quarter (in \$k): Q1=50, Q2=65, Q3=55, Q4=80. a) Compute average quarterly sales. b) What percent larger is Q4 than the average? (Round to one decimal.)

Answer Key (brief)



1. Total person-days = $6 * 8 = 48$. With 4 people: days = $48 / 4 = 12$ days.
2. Simple interest = $P * r * t = 5000 * 0.04 * 3 = \600 .
3. Increase = $150 - 120 = 30$. Percent increase = $30 / 120 = 0.25 = 25\%$.
4. Ratio parts = $3 + 2 = 5$ parts. A = $3/5 * 25 = 15$ L. B = $2/5 * 25 = 10$ L.
5. $3(x - 2) + 4 = 2(x + 5) - x$
 $3x - 6 + 4 = 2x + 10 - x$
 $3x - 2 = x + 10$
 $2x = 12 \rightarrow x = 6$.
6. Differences: 4, 6, 8, 10 $\rightarrow$ increases by even numbers. Next add 12: $30 + 12 = 42$.
7. Red or green = $(5 + 2) / (5 + 3 + 2) = 7/10 = 0.7 = 70\%$.
8. Discounted price = $250 * (1 - 0.20) = 200$. After tax = $200 * (1 + 0.08) = 216$. Final = \$216.
9. Let width = w , length = $2w$. Perimeter = $2(2w + w) = 6w = 60 \rightarrow w = 10$ cm. Area = length * width = $20 * 10 = 200 \text{ cm}^2$.
- 10a) Average = $(50 + 65 + 55 + 80) / 4 = 250 / 4 = 62.5$ (\$k). 10b) Q4 compared to average = $(80 - 62.5) / 62.5 = 17.5 / 62.5 = 0.28 = 28.0\%$.



Coding Problem — Algorithmic Exercise

Problem: Longest Consecutive Sequence

·Time: 30 minutes

·Difficulty: Medium

·Goal: Given an unsorted array of integers, return the length of the longest sequence of consecutive integers.

·Requirements: Provide a clear, working solution (C# preferred). State time and space complexity and any assumptions.

Input / Output

·Input: integer array `nums[]`

·Output: single integer (length of longest consecutive sequence)

Examples

·`nums = [100, 4, 200, 1, 3, 2]` -> 4 (sequence 1,2,3,4)

·`nums = [0, -1, 1, 2]` -> 4 (sequence -1,0,1,2)

·`nums = []` -> 0

Constraints

· $0 \leq \text{nums.length} \leq 100,000$

·Values fit in 32-bit signed int

·Expected: $O(n)$ average time, $O(n)$ extra space

Grading rubric (suggested)

·Correctness on edge and large inputs (50%)

·Time complexity meets $O(n)$ average (25%)

·Code clarity, comments, and tests (25%)

Hints

·Use a `HashSet` for $O(1)$ membership checks.



· For each number, start a forward scan only if $(\text{num} - 1)$ is not present to avoid repeated work.

Pseudocode

1. Put all nums into a set S.
2. For each n in S:
 - if $n-1$ not in S:
 - length = 1
 - while $n+\text{length}$ in S: length++
 - update maxLen
3. Return maxLen



Excel / Data Skills Test — 20 minutes

Instructions

- Time: 20 minutes. Show steps and include formulas or screenshots where requested.
- Work on a copy of the dataset. Use Excel, Google Sheets, or similar.
- Provide formulas, brief explanation, and expected outputs.

Dataset (first 8 rows; Date in YYYY-MM-DD, Amount in USD)

Date,Salesperson,Region,Amount

2025-01-05,Alice,North,1200

2025-01-12,Bob,South,850

2025-02-02,Alice,North,950

2025-02-15,Charlie,East,700

2025-03-01,Bob,South,1100

2025-03-10,Alice,North,1300

2025-03-22,Charlie,East,400

2025-04-05,Bob,South,900

Tasks

1.Pivot / aggregation (5 min)

- Create a pivot or use formulas to show total Amount by Region and by Salesperson within each Region.
- Deliverable: screenshot or exported table.

2.Rolling / moving average (6 min)

- Compute a 3-month rolling average of monthly total Amounts (month based on Date).
- Provide the formula used (preferably using Excel functions: AVERAGEIFS, SUMIFS+OFFSET, or dynamic functions like LET / AVERAGE if available).
- Example expected results (monthly totals then 3-month rolling averages).



3. Flag outliers (4 min)

- Add a column that flags Amounts that are outliers based on IQR (mark "Outlier" if Amount $> Q3 + 1.5 \cdot IQR$ or $< Q1 - 1.5 \cdot IQR$).
- Provide the formula used to compute Q1, Q3, IQR and the flag.

4. Quick analysis question (5 min)

- Which salesperson had the highest total sales in the dataset? Show the formula or pivot used.
- Provide one recommendation (1–2 sentences) for investigating a sudden drop or spike in a salesperson's monthly totals.

Sample solutions / formulas (concise)

4. Pivot: Rows = Region, Salesperson; Values = Sum of Amount.
5. 3-month rolling average (assuming monthly totals table with Month in A2 and Total in B2):
 - Excel (pre-Dynamic): in C4: **=AVERAGE(B2:B4)** and copy down to align windows.
 - Dynamic months (if dates per row): monthly total with **SUMIFS(AmountRange, DateRange, ">="&EOMONTH(thisMonth, -2)+1, DateRange, "<="&EOMONTH(thisMonth, 0)) / 3**
6. IQR outlier flag:
 - Q1: **=QUARTILE.INC(AmountRange, 1)**
 - Q3: **=QUARTILE.INC(AmountRange, 3)**
 - IQR: **=Q3 - Q1**
 - Flag (for Amount in cell D2): **=IF(OR(D2 > Q3 + 1.5*IQR, D2 < Q1 - 1.5*IQR), "Outlier", "")**
7. Highest total salesperson:
 - Pivot result or formula: **=INDEX(SalespersonRange, MATCH(MAX(SalesTotalsRange), SalesTotalsRange, 0))**



- Recommendation example: "Check transaction-level data and date-based filters for that month; verify returns/refunds and confirm sales credits weren't posted to another account."

Grading guidance (optional)

- Correct pivot/aggregation: 30%
- Correct rolling average calculation: 30%
- Correct outlier identification and formulas: 25%
- Clear explanations and recommendations: 15%



Customer Service Scenarios Test — 20 minutes

Instructions

- Time: 20 minutes. Answer all scenarios using the STAR method (Situation, Task, Action, Result).
- Write concise, structured responses (recommended: 3–6 sentences per STAR section).
- Score each response against the rubric below.

Scenarios (answer each)

1.Repeated Billing Error (5 min)

- Situation: A long-term client calls, frustrated about recurring incorrect invoices and requests an immediate refund. They threaten to take their business elsewhere.
- Prompt: Using STAR, explain how you would handle the call, steps to resolve the billing issue, how you'd communicate timelines, and measures to prevent recurrence.

2.Missed Deadline, Escalated Customer (5 min)

- Situation: A key deliverable missed its SLA. The customer has escalated to management and demands priority resolution and compensation.
- Prompt: Describe how you would de-escalate, coordinate internal teams to fix the issue, set expectations with the customer, and propose fair remediation.

3.Technical Confusion — Non-technical Customer (4 min)

- Situation: A non-technical customer cannot use a new product feature and is confused by technical error messages.
- Prompt: Show how you'd explain the problem in plain language, guide them through troubleshooting, and ensure they understand next steps.

4.High-Value Retention Call (6 min)

- Situation: A major client indicates they're considering a competitor due to perceived lack of responsiveness over the last quarter.
- Prompt: Outline a retention plan: immediate actions on the call, short-term remediation, tactical changes, and metrics to demonstrate improvement.



Scoring rubric (per scenario)

- Empathy & tone (0–3): Acknowledges frustration, uses calming language.
- Clear action plan (0–4): Concrete next steps, owner identified, reasonable timelines.
- Problem ownership & escalation (0–2): Appropriate escalation and follow-up commitments.
- Preventive measures (0–1): Shows how to stop recurrence. Total per scenario: 0–10 points.

Sample STAR guidance (brief)

- Situation: One sentence summarizing context.
- Task: What you needed to achieve for the customer.
- Action: Step-by-step actions you took (include who you involved, what you checked, what you promised).
- Result: Outcome, expected timeline, follow-up, and how recurrence was prevented.

Example (short)

- Situation: Repeated incorrect invoices to a long-term client.
- Task: Restore accuracy, issue any necessary refunds, and restore trust.
- Action: Apologize, validate account and invoices live on the call, escalate to billing with priority ticket and expected 48-hour turnaround, issue provisional credit if policy allows, and schedule written follow-up.
- Result: Client received correction within 48 hours, billing root cause fixed (misconfigured tax code), and client accepted goodwill credit — reduced errors by process change.

Evaluation tips for graders

- Look for explicit timelines and ownership.
- Prefer measurable remediation (e.g., “refund issued within 48 hours”, “follow-up email within 24 hours”).
- Give extra credit for proposals that reduce recurrence (process, monitoring, checklist, automation).